

Online Supplement

Source-Scored Structural Initial Designs for Chance-Constrained Optimization of Ordered Policy Profiles

Authors' names are not included for peer review

EC.1. Finite Algorithm

Algorithm 1: Source-scored structural initial design

1. Generate the declared 64-profile outcome-free library on 128 regular midpoints using seed 20260808.
2. For every source task and profile, compute the replicated objective mean and the Gaussian chance-margin statistic in (8).
3. Convert each statistic to deterministic within-task average-tie percentile ranks; average the ranks uniformly across source tasks.
4. Compute exact Voronoi-cell cosine coefficients $c_0, \dots, c_8$, divide c_k by $1 + .25k$, append all nine diagonal squares, and standardize the 18 coordinates over the library.
5. Append source safety and objective ranks. Select the first center by (11); select nine further centers by (12).
6. Linearly interpolate each selected profile to the declared target grid, clip to bounds, and deterministically round. Freeze these ten policies before the first target outcome.
7. Run any declared target backend. Freeze a distinct ordered shortlist before verification.
8. Test each shortlist candidate on its independent all-success stream. Deploy the first certificate; abstain if none certifies.

The stable tie order is safety rank, objective rank, profile identifier for the first center and finite-library index for farthest-first. Robust source feasibility is diagnostic only. There is no endpoint replacement, hidden scalarization, target-outcome repair, or target-oracle branch in a non-oracle arm.

EC.2. Additional Statistical Results

EC.2.1. Verifier thresholds

Table EC.1 gives exact power calculations. At $n = 80$, the Clopper–Pearson lower-bound rule equals the all-success rule. At $n = 160$, its preregistered sensitivity threshold permits two failures. The primary results retain the all-success protocol.

EC.2.2. Full sensitivity matrix

The full one-factor sensitivity matrix is reported in Table EC.2. Entries are ordered by the tested levels in the second column. Effective-rank levels are explicit overrides; all other rows include

Table EC.1 Exact candidate-wise certificate power at budgets 80 and 160.

n	p	All-success power	CP threshold	Allowed failures	CP power
80	0.950	0.017	80	0	0.017
80	0.975	0.132	80	0	0.132
80	0.990	0.448	80	0	0.448
80	0.995	0.670	80	0	0.670
80	1.000	1.000	80	0	1.000
160	0.950	0.000	158	2	0.012
160	0.975	0.017	158	2	0.234
160	0.990	0.200	158	2	0.784
160	0.995	0.448	158	2	0.953
160	1.000	1.000	158	2	1.000

the frozen baseline level. The nonmonotone responses, the zero certificate rate at $\alpha = .01$, and the immaterial frequency penalty are retained rather than optimized away.

Table EC.2 One-factor sensitivity of the source-scored design at $d = 1000$. Each rate aggregates eight fixed regime strata and 20 independently seeded latent tasks per regime at one resolution.

Factor	Tested levels	Feasible (%)	Certified (%)
Effective rank	4/8/16/32	89.4/85.0/84.4/83.8	55.6/36.3/25.6/29.4
Chance level α	.01/.05/.1	95.0/95.0/95.0	0.0/46.2/29.4
Calibrated safe mass	.03/.08/.18	69.4/95.0/99.4	22.5/46.2/70.0
Initial-design size n_0	5/10/20	79.4/95.0/96.9	42.5/46.2/46.9
Source tasks	1/2/4	85.0/95.0/95.6	33.8/46.2/45.0
Profiles per source task	32/64/128	83.1/95.0/73.8	38.1/46.2/38.8
Source replications	1/3/5	92.5/95.0/92.5	45.0/46.2/45.6
Retained frequency K	4/8/16	81.9/95.0/95.6	43.1/46.2/48.1
Frequency penalty κ	0/.25/1	95.0/95.0/95.0	46.2/46.2/46.2
Safety-rank weight	.25/1/4	95.0/95.0/95.0	46.9/46.2/46.2
Objective-rank weight	.25/1/4	95.0/95.0/95.0	46.9/46.2/46.2
First-center safety weight	.25/.5/.75	62.5/95.0/95.0	32.5/46.2/45.0

Recorded wall time is descriptive rather than a resource-normalized endpoint. On the mixed CPU cluster, the primary ten-point initial-design arms average 1.2–1.7 seconds per synthetic task, whereas 394-call source-free controls average about 59 seconds and the 394-call functional-SCBO arm has a median of about 677 seconds. The synthetic simulator is analytic and cheap, so these figures measure implementation overhead rather than the time of a physical simulator; simulation-call accounting remains the portable comparison.

EC.2.3. Task-seed strata audit

The registered generator includes a deterministic `task_seed mod 5` cosine component. Table EC.3 reports its realized categories rather than silently treating the finite sample as a balanced IID mixture. The analysis uses fixed regime strata and applies concentration separately at each resolution.

Table EC.3 Realized latent-category counts among the 160 independent task seeds. The same seeds are crossed with all three grid resolutions.

Regime	Category 0	1	2	3	4	Total
Aligned low frequency	3	8	3	1	5	20
Growing rank	4	7	6	3	0	20
Frequency shift	8	2	3	2	5	20
Permutation	11	0	3	5	1	20
Irregular grid	4	4	6	4	2	20
Piecewise smooth	5	4	6	4	1	20
Sparse high frequency	3	1	7	4	5	20
Misspecified target	3	6	6	2	3	20
Overall	41	32	40	25	22	160

EC.2.4. Native end-to-end transfer

Table EC.4 Native transfer pipelines at $N = 20$ with the same 384-call source archive and independent verifier. Each method retains its own initialization and online adaptation.

Native method	FactorShock	Inventory	Queue	Total	False
Safe F-PACOH	0/20	0/20	0/20	0/60	0
RGPE	14/20	16/20	0/20	30/60	0
Stacked GP	0/20	13/20	0/20	13/60	0
MTGP	1/20	0/20	0/20	1/60	0
FSBO	6/20	2/20	0/20	8/60	0
HyperBO	0/20	0/20	0/20	0/60	0
MetaBO	19/20	9/20	0/20	28/60	0
MALIBO	12/20	2/20	0/20	14/60	0

No method certifies a Queue task. These 60-task totals measure repeatability on three fixed families, not generalization to a population of domains.

EC.2.5. Energy temporal audit

Table EC.5 Postdecision Energy temporal stability. Counts are descriptive and do not replace the declared empirical-window certificate.

Matrix	Originally certified	Block stable	Nonoverlap stable	Joint stable
Energy V2	317	303/317	316/317	303/317
Functional V2	62	59/62	61/62	58/62
Energy V3	358	312/358	358/358	312/358

V3 preserves all 358 original certificates under physically nonoverlapping windows but 312 under the joint block criterion. Shared calendars and only five regions preclude an iid market-population interpretation.

EC.3. Secondary Cumulative-Variance Diagnostic

The project originally studied cumulative heteroscedastic variance decomposition. Under matched policies, the factor-HVD model represents local and shared exposure and improves held-out variance-shape estimation. Table EC.6 gives the final matched diagnostic.

Table EC.6 Pooled variance versus cumulative factor-HVD. HVD improves calibration but not feasible recovery or verification cost.

Domain	Variance model	Feasible	Log-RMSE	Shape corr.	Mean verify
FactorShock	Pooled	20/20	1.158	0.000	159.2
FactorShock	Cumulative factor-HVD	20/20	0.219	0.919	158.4
Inventory	Pooled	20/20	0.577	0.000	182.4
Inventory	Cumulative factor-HVD	20/20	0.349	0.953	184.0
Queue	Pooled	19/20	0.501	0.000	190.4
Queue	Cumulative factor-HVD	19/20	0.254	0.995	197.6

Because both variance models recover the same feasible counts and HVD slightly increases verification cost in Inventory and Queue, HVD is not included in the paper's claimed optimization method. Its formal results remain in the proof archive as reusable mechanistic results.

EC.4. Human-Readable Proofs

This section gives the complete conventional proof chain used by the paper. It includes the two short geometric arguments rather than leaving them as verbal sketches. Section EC.5 then maps every step to an exact Lean declaration and separates proved implications from scientific premises.

EC.4.1. Profile-coordinate stability

Proof of Proposition 1. Let $e_d = h - \widehat{h}_d$. Since the integration interval has length one, the triangle inequality gives

$$\begin{aligned} |c_k(h) - c_k(\widehat{h}_d)| &= \left| \int_0^1 e_d(t) b_k(t) dt \right| \\ &\leq \int_0^1 |e_d(t)| |b_k(t)| dt \leq \epsilon_d B_k. \end{aligned}$$

For the regular-midpoint reconstruction, every t lies within $1/(2d)$ of the midpoint assigned to its Voronoi cell. Lipschitz continuity therefore implies

$$|h(t) - \widehat{h}_d(t)| \leq L_h/(2d),$$

so substituting $\epsilon_d = L_h/(2d)$ proves the first rate. Dividing both coefficients by $1 + \kappa k \geq 1$ divides their absolute difference by the same quantity and cannot enlarge it.

For completeness, suppose t lies between adjacent reference nodes l and r , has interpolation weight $\theta \in [0, 1]$, and the node spacing is $1/d_{\text{ref}}$. The interpolant is $\widehat{h}(t) = (1 - \theta)h(l) + \theta h(r)$, whence

$$\begin{aligned} |h(t) - \widehat{h}(t)| &\leq L_h\{(1 - \theta)|t - l| + \theta|t - r|\} \\ &= \frac{2L_h\theta(1 - \theta)}{d_{\text{ref}}} \leq \frac{L_h}{2d_{\text{ref}}}. \end{aligned}$$

This is the claimed pre-rounding interpolation bound.

EC.4.2. Farthest-first coverage

Proof of Theorem 1. Let $r = r(A_k; u)$ and choose a library point w whose distance to A_k is r . Farthest-first selection implies that the k selected centers together with w are pairwise separated by at least r : when each center was inserted, its distance from all earlier centers was at least the final covering radius, and the same is true of w by construction.

Assign these $k + 1$ witnesses to their nearest centers in an optimal k -center solution of radius $r_k^\star$. Two witnesses, say x and y , must share one optimal center o . The triangle inequality then yields

$$r \leq \|u(x) - u(y)\|_2 \leq \|u(x) - u(o)\|_2 + \|u(o) - u(y)\|_2 \leq 2r_k^\star.$$

The zero-radius case is immediate. Finally, deleting the two source-rank coordinates is a Euclidean projection and is nonexpansive. Hence an $\eta = (z, r^g, r^f)$ cover of radius r is also a z cover of radius at most r , proving Equation (14).

EC.4.3. Finite source errors and separated-rank recovery

Proof of Proposition 2 and the finite-source margin bound. Fix a source task and a library profile. For its R independent Gaussian constraint replications, write $\bar{Y}$ and S for the sample mean and sample standard deviation, and write μ and σ for their population counterparts. The Gaussian mean tail gives

$$\mathbb{P}\{|\bar{Y} - \mu| > r_\mu(\delta_\mu)\} \leq \delta_\mu.$$

Let $\nu = R - 1$. Cochran's identity gives $(R - 1)S^2/\sigma^2 \sim \chi_\nu^2$, and therefore, with probability at least $1 - \delta_\sigma$,

$$\sqrt{q_{\nu, \delta_\sigma/2}/\nu} \leq \frac{S}{\sigma} \leq \sqrt{q_{\nu, 1-\delta_\sigma/2}/\nu}.$$

On this event,

$$|S - \sigma| \leq \sigma \max\{a_\nu(\delta_\sigma), b_\nu(\delta_\sigma)\} = r_\sigma(\delta_\sigma).$$

Because $\sigma \geq 10^{-3}$, replacing S by $\max\{S, 10^{-3}\}$ cannot increase its distance from σ . Hence, on the intersection of the mean and scale events,

$$\begin{aligned} |\hat{m} - m| &= |(\bar{Y} - \mu) + z_{1-\alpha}(\max\{S, 10^{-3}\} - \sigma)| \\ &\leq r_\mu(\delta_\mu) + z_{1-\alpha}r_\sigma(\delta_\sigma). \end{aligned}$$

The objective score uses the same argument without the scale term. Assigning the tail probabilities across the finite set of source tasks, profiles, and score heads and applying the union bound produces a simultaneous radius r_q for each score head.

It remains to show what this event implies for ranking. Let profiles j and ℓ have true scores $s_{qj} < s_{q\ell}$ with $s_{q\ell} - s_{qj} > 2r_q$. Uniform error at most r_q gives

$$\hat{s}_{qj} \leq s_{qj} + r_q < s_{q\ell} - r_q \leq \hat{s}_{q\ell}.$$

Thus every pair separated by more than $2r_q$ has the correct estimated order. Pairs inside that band may remain tied or reverse, which is why the result does not claim exact recovery of the full percentile ranking.

EC.4.4. From structural coverage to a feasible target policy

Proof of Theorem 2. By Assumption 1, choose a source-supported library profile h_s satisfying $d_q(h_s, h_q^\star) \leq \Delta_q$. Because the selected atlas covers the library in augmented coordinate η , there is an atlas member a such that

$$\|\eta(a) - \eta(h_s)\|_2 \leq r_{n_0}^\eta.$$

Projection onto the structural coordinates is nonexpansive, so

$$\|\widehat{z}(a) - \widehat{z}(h_s)\|_2 \leq r_{n_0}^\eta.$$

Insert $\widehat{z}(a)$, $\widehat{z}(h_s)$, and $z_q^\star(h_s)$ between the ideal target coordinates of a and the safe center. The triangle inequality and the two coordinate-approximation bounds give

$$\begin{aligned} d_q(a, h_q^\star) &\leq \|z_q^\star(a) - \widehat{z}(a)\|_2 + \|\widehat{z}(a) - \widehat{z}(h_s)\|_2 \\ &\quad + \|\widehat{z}(h_s) - z_q^\star(h_s)\|_2 + d_q(h_s, h_q^\star) \\ &\leq r_{n_0}^\eta + \Delta_q + 2\epsilon_\eta. \end{aligned}$$

The one-sided Lipschitz condition for the chance margin now yields

$$\begin{aligned} m_q(a) &\leq m_q(h_q^\star) + L_q d_q(a, h_q^\star) \\ &\leq -\gamma_q + L_q(r_{n_0}^\eta + \Delta_q + 2\epsilon_\eta) \leq 0, \end{aligned}$$

where the last inequality is exactly condition (20). Therefore a is target feasible and belongs to the selected atlas.

EC.4.5. Stratified task-law concentration

Proof of Equation (21). Index the independent held-out task outcomes by (s, i) and write $a_{si} = w_s/n_s$. Then

$$\widehat{p} - p = \sum_s \sum_{i=1}^{n_s} a_{si} \{H_{si} - \mathbb{E}[H_{si}]\}.$$

The variables need not have the same distribution. Independence and $H_{si} \in [0, 1]$ are enough for Hoeffding's lemma, which gives

$$\mathbb{E} \exp\{\lambda(\widehat{p} - p)\} \leq \exp\left\{\frac{\lambda^2}{8} \sum_s \sum_{i=1}^{n_s} a_{si}^2\right\} = \exp\{\lambda^2 v_w / 8\},$$

because $\sum_i a_{si}^2 = w_s^2 / n_s$. Chernoff optimization at $\lambda = 4r / v_w$ gives

$$\mathbb{P}\{\widehat{p} - p \geq r\} \leq e^{-2r^2 / v_w}.$$

Applying the same argument to $p - \widehat{p}$ and taking a union bound yields $\mathbb{P}\{|\widehat{p} - p| \geq r\} \leq 2e^{-2r^2 / v_w}$.

Set

$$r = \sqrt{\frac{v_w \log(2/\delta)}{2}}.$$

The right-hand side is δ , and on the complementary event $p \geq \widehat{p} - r$, which is Equation (21). Applying the argument separately at each resolution is valid because independence is required within a resolution, not between the three reported resolutions.

EC.4.6. Exact familywise verification

Proof of Theorem 3. Consider a fixed candidate j whose true success probability satisfies $p_j < p_0$. Certification requires all v_j fresh Bernoulli trials to succeed. Independence within that candidate's verification stream gives

$$\mathbb{P}\{j \text{ certifies}\} = p_j^{v_j} \leq p_0^{v_j} \leq \delta_j.$$

If an unsafe policy is deployed from the fixed shortlist, at least one unsafe shortlist member must have certified. Consequently,

$$\mathbb{P}\{\text{an unsafe policy is deployed}\} \leq \sum_{j: p_j < p_0} \mathbb{P}\{j \text{ certifies}\} \leq \sum_{j=1}^{K_s} \delta_j \leq \delta.$$

This union bound does not require independence between candidates. Testing them in a fixed order and stopping at the first certificate only removes later tests from consideration; it does not enlarge the event that any unsafe candidate passes. Fixing the shortlist, its order, and all budgets before verification is what prevents search or selection from using the fresh test outcomes.

EC.4.7. Source-only scope, no-free-lunch, and call accounting

The frozen proposal is a function of source records and the declared target grid only. Replacing one unused target-label object by another therefore leaves the proposal unchanged. This is a definitional noninterference result, not a probabilistic generalization claim.

For the corresponding limitation, let $\mathcal{X}$ be finite and let a deterministic target-label-free atlas A be a proper subset of $\mathcal{X}$. Choose $x \in \mathcal{X} \setminus A$ and define the nonempty feasible set as $\mathcal{F} = \{x\}$. Then $A \cap \mathcal{F} = \emptyset$. In particular, $|A| \leq n_0 < |\mathcal{X}|$ implies that such a feasible set exists. Thus a finite source-only proposal needs the alignment premise in Theorem 2; coverage of every possible target is impossible.

Finally, if the source archive costs S simulations, target search costs N , and independent verification costs V , disjoint accounting gives $C_{\text{all}} = S + N + V$. If the same frozen archive is reused by M target problems, charging each target an equal share of the archive gives $C_{\text{amort}}(M) = S/M + N + V$. These are accounting identities; they do not claim that archive reuse is statistically valid for an unrelated target.

EC.5. Machine-Checked Proof Map

The Lean 4 project builds against mathlib without `sorry`, `admit`, or a project-defined axiom. The authoritative paper-facing manifest is `proof/SCOLHKG/PaperProofInterface.lean`. It contains compile-time checks for exactly the declarations in Table EC.7. Every conventional proof in Section EC.4 maps to at least one declaration, and every declaration in the manifest maps back to a manuscript statement or accounting identity. Other files in the larger Lean archive formalize historical KG, HVD, and intermediate algorithm versions; they are not claims of this paper.

The map distinguishes logical closure from empirical truth. Assumption 1 is intentionally a premise consumed by the aligned-coverage theorem, not a theorem about every target. For replicated Gaussian source scores, the normal mean law and Cochran's chi-square identity are standard distributional inputs; Lean verifies the finite-sample variance identity, event union, variance-floor propagation, and rank implication after those inputs are supplied. Likewise, the task-law result assumes independent bounded held-out tasks, and terminal verification assumes the declared binomial law for each fresh stream. These boundaries are part of the theorem statements rather than unrecorded gaps.

Table EC.7 Bidirectional closure of manuscript statements, conventional proofs, and the final paper-facing Lean interface.

Names after the file colon are exact Lean declarations.

Manuscript statement	Conventional proof	Lean file and declaration
Target-label noninterference	Section EC.4.7	Real/RiskAlignedRepresentation.lean: sourceOnlyProposal_targetLabel_invariant
Proposition 1 and interpolation rate	Section EC.4.1	Real/ProfileCoordinateConsistency.lean: continuousProfileCoefficient_error_le; lipschitz_voronoi_coefficient_inverse_grid_rate; lipschitz_linear_interpolation_inverse_grid_rate; frequency_penalty_cannot_increase_coefficient_error
Theorem 1 and projected cover	Section EC.4.2	Real/FarthestFirstKCenter.lean: GonzalezWitnessCertificate.two_approx_every_k_center; Real/GeometricAtlasCoverage.lean: projected_coordinate_atlas_covers
Proposition 2 and finite margin	Section EC.4.3	Measure/SourceRankRecovery.lean: finite_source_profile_mean_bad_event_le_sum; floored_margin_bad_event_measure_le; Real/SourceRankRecovery.lean: floored_empirical_chance_margin_error_le; separated_source_pair_order_recovered
Theorem 2	Section EC.4.4	Real/GeometricAtlasCoverage.lean: finite_projected_aligned_geometric_lipschitz_atlas_coverage
Equation (21)	Section EC.4.5	Measure/TaskAtlasCoverage.lean: weightedTaskCoverageError_subGaussian; weightedTaskCoverageError_abs_tail_le
Theorem 3	Section EC.4.6	Measure/ExactBinomialCertificate.lean: unsafe_all_success_certificate_probability_le; finite_all_success_certificates_familywise
Finite-atlas limitation	Section EC.4.7	Real/ProposalNoFreeLunch.lean: proper_finite_atlas_misses_some_nonempty_feasible_set; finite_budget_no_unconditional_target_coverage
Unamortized and amortized call identities	Section EC.4.7	Real/MethodIndependentTerminalVerification.lean: paper_grade_budget_exact_decomposition; paper_grade_budget_amortized_decomposition

EC.6. Reproducibility Materials

A frozen evidence registry links 17,360 result cells to 14 experimental matrices and 12 compact analyses. It retains the one recorded algorithmic failure. Every row identifies the task, source archive, initial design, method, implementation version, verifier, source/search/verification calls, and the truth used only after the decision.

Tables and figures in this manuscript are generated only from compact JSON and CSV analyses. The renderer does not read checkpoints, pickle files, model weights, or exported policy vectors. Runtime artifacts are unnecessary for auditing the displayed results. The method specification, randomized task law, Energy data specification, proof receipt, and rendering manifest are versioned separately so that results from one method version cannot be paired with theory from another. The geometry, task-strata, and cost analyses use the same frozen evidence registry as the primary outcomes.